Arista Access Node BAS-ACC-78 Basavanagudi POP QoS Congestion Event SOP

Node: BAS-ACC-78 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_5697

Description

This SOP addresses a critical failure scenario on Arista's Access Node BAS-ACC-78 located at the Basavanagudi POP, where a QoS congestion event is triggered by microbursts, interface tail drops, or QoS queue starvation. The root cause of this failure lies in the interplay between the Access node's egress queue management, the buffer size configuration, and the QoS policies in place. Specifically, the failure occurs when the egress queue becomes congested due to an imbalance in traffic distribution, causing tail drops and subsequent QoS queue starvation. This congestion event can be exacerbated by an inadequate buffer size configuration, which fails to absorb the bursty traffic, leading to a cascading effect on the network performance. Understanding the intricate relationship between the egress queue management, buffer size configuration, and QoS policies is crucial in resolving this failure scenario.

Pre-requisite Checks

- Ensure remote hands are dispatched to the Basavanagudi POP to verify the link status and confirm that the redundant link is functioning correctly.

- Verify that the buffer size configuration on the egress queue is adequate to handle the expected bursty traffic.

- Confirm that the QoS policies in place are not causing an imbalance in traffic distribution, leading to congestion on the egress queue.

Potential Root Causes

Cause	Probability
Inadequate buffer size configuration	High
Insufficient egress queue size	Medium
QoS policy imbalance leading to traffic congestion	Low

Diagnostic Commands & Expected Outputs

```
Command: show queue egress 1/1 detail, Expected Output: Buffer size: 8192 bytes, Threshold: 50%, Description: This command displays the egress queue details, including the buffer size and threshold values. The buffer size should be set to at least 8192 bytes to handle the expected bursty traffic.

Command: show qos policy 1/1, Expected Output: QoS policy: 'default qos policy', Class of Service: '0x0', Description: This command displays the QoS policy in place, including the Class of Service (CoS)
```

```
value. The CoS value should be set to '0x1' to prioritize traffic
based on CoS.

Command: show buffer utilization 1/1, Expected Output: Buffer
utilization: 75%, Description: This command displays the buffer
utilization on the egress queue. The buffer utilization should be
below 50% to prevent congestion.
```

Resolution Steps

Step 1: Step: Increase egress queue buffer size, Command: queue egress 1/1 buffer-size 16384, Description: Increase the buffer size on the egress queue to 16384 bytes to handle the expected bursty traffic.

Step 2: Step: Adjust QoS policy to prioritize CoS 1 traffic, Command: qos policy 1/1 class-of-service 0x1, Description: Adjust the QoS policy to prioritize traffic based on CoS 1, ensuring that critical traffic is not affected by the congestion event.

Step 3: Step: Verify buffer utilization is below 50%, Command: show buffer utilization 1/1, Description: Verify that the buffer utilization on the egress queue is below 50% to prevent congestion.

Post-Check Verification

```
Command: show queue egress 1/1 detail, Expected Output: Buffer size:
16384 bytes, Threshold: 50%, Description: Verify that the egress queue
buffer size has been increased to 16384 bytes and the threshold is set
to 50%.

Command: show qos policy 1/1, Expected Output: QoS policy: 'default
qos policy', Class of Service: '0x1', Description: Verify that the QoS
policy has been adjusted to prioritize CoS 1 traffic.

Command: show buffer utilization 1/1, Expected Output: Buffer
utilization: 25%, Description: Verify that the buffer utilization on
the egress queue is below 50% and the congestion event has been
mitigated.
```

Roll-back Procedures

- Step: Restore original egress queue buffer size, Command: queue egress 1/1 buffer-size 8192, Description: Restore the original buffer size configuration on the egress queue to prevent any adverse effects on the network performance.

- Step: Restore original QoS policy, Command: qos policy 1/1 class-of-service 0x0, Description: Restore the original QoS policy to prevent any adverse effects on the network performance.